Cade Rousseau

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SUMMARY

B.S. Software Engineering student at MSOE with a minor in Data Science. High-performing developer with a focus on enterprise AI integration, full-stack systems, and open-source software development. Proven ability to deliver scalable technical solutions in fast-paced corporate environments, with a track record of building secure agentic AI architecture, optimizing complex data pipelines, and designing high-performance telemetry tools. Technical expertise spans Java, JavaScript, ML, Python, C++, React, and enterprise software design. Eager to leverage strong debugging, architectural modeling, and systems engineering skills in a full-time capacity starting May 2027.

EDUCATION

B.S. Software Engineering, Minor in Data Science | Milwaukee School of Engineering | GPA: 3.6 | May 2027

INTERSHIP EXPERIENCE

AI Intern – IBM
May 2026 – August 2026 | Rochester, MN

- Architected a lightweight, enterprise-grade Model Context Protocol (MCP) server using Java 17 to enable secure deployment of autonomous AI agents in highly regulated industries.
- Engineered secure agentic workflows by eliminating standard open-source Node.js dependency stacks, ensuring strict compliance for enterprise clients in banking and healthcare.
- Streamlined the evaluation of a novel agentic AI product by coordinating testing phases and gathering critical performance data for private client previews.
- Visualized enterprise integration capabilities by building secure, localized AI tools, transforming raw protocol specifications into production-ready open-source components.
- Optimized product readiness through rigorous validation workflows during private previews, accelerating the feedback loop between core engineering and business clients.

Software Engineering Intern – GE Healthcare (via Codeworks)
September 2025 – May 2026 | Waukesha, WI

- Architected a full-stack internal dashboard using Node.js and JavaScript to visualize complex call trees and filter production logs, enhancing system transparency for engineering teams.
- Engineered asynchronous event correlation by identifying and implementing logging fixes across production microservices, enabling the tracing of fragmented system events.
- Streamlined log analysis using AI-assisted tools to rapidly parse thousands of daily events, accelerating the identification of system anomalies and performance bottlenecks
- Visualized system parallelism by mapping synchronous and asynchronous calls on Linux-based hardware, transforming raw telemetry into actionable performance insights.
- Optimized debugging workflows through the design of real-time monitoring interfaces, significantly reducing the time required to validate backend microservice interactions.

PROJECT EXPERIENCE

- **ROSIE Supercomputer Dashboard (Group of 4)**
  - Full-Stack Development: Built a Next.js/React visualizer to display real-time SLURM job activity and per-node telemetry (CPU, GPU, Network) to be displayed at Diercks Hall at MSOE.
  - Systems Integration: Maintained and optimized a Node.js/Express API that aggregates data from UDP streams, MariaDB, and SSH into unified REST endpoints.
  - DevOps & Testing: Modernized deployment using GitLab CI/CD and Docker; implemented Jest suites to ensure stability across the legacy and updated codebase.

- **FISH Image Analysis (Group of 4)**
  - Partnering with Mayo Clinic on a semester-long practicum to automate Fluorescence In Situ Hybridization (FISH) image analysis through two deep-learning pipelines: a binary validity classifier (Task 1) and a dual-head probe counter (Task 2).
  - Built a hybrid EfficientNetB1 + U-Net model in TensorFlow/Keras for Task 1, using Binary Focal Cross-Entropy with class weighting and a two-phase frozen-then-fine-tuned training schedule, achieving an F1 score of 0.91 on Marked vs. Rejected classification.
  - Built a DINOv3 ViT-S/16 backbone with dual CORAL ordinal heads in PyTorch for Task 2 to count red and green probes, applying LDAM margins, weighted sampling, and validation-calibrated thresholds to handle class imbalance, achieving a combined F1 score of 0.96 across 5-class classification.
  - Trained and evaluated on a SLURM-managed multi-GPU HPC cluster, delivering reproducible models, predictions, and performance reports for clinical review.
- **Brain Aneurysm Detection & Segmentation (Group of 4)**
  - Developed a two-stage 3D deep learning pipeline for automated aneurysm detection in CTA scans as part of an MSOE AI Club initiative and a global Kaggle competition.
  - Orchestrated a deterministic geometry pipeline to standardize over 1 million DICOM slices into unified 3D volumes, utilizing 13 arterial masks to focus model training on high-relevance vascular structures.
  - Leveraged the ROSIE Supercomputer to perform progressive fine-tuning on NVIDIA H100 GPUs, achieving an 89.6% detection sensitivity and a 59% improvement in composite score over the baseline.
  - Selected to present research findings and methodology at MICS 2026 and the MSOE Rosie Competition poster session.

TECHNICAL SKILLS

- **Languages/Frameworks**
  - JavaScript, HTML, CSS, TypeScript, Java, Python, C/C++, React, JavaFX, REST APIs, SQL
- **Databases**
  - MongoDB, Relational Databases (SQL)
- **Tools/Tech:**
  - AWS (Amplify), Git/GitHub, Agile/Scrum, Visual Studio Code, UML Modeling, Design Patterns
- **Libraries:**
  - Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow
- **Concepts:**
  - UI/UX Principles, Web Accessibility Standards, Debugging, Version Control
- **Certifications:**
  - Fundamentals of Deep Learning – NVIDIA DLI

LEADERSHIP | CO-CURRICULAR INVOLVEMENT | COMMUNITY SERVICE

Cadet | Army ROTC | January 2024 – December 2024 | 20 hrs. per wk.
Member | Society of Software Engineers | | August 2023 - Present| 1 hrs. per wk.
Member | AI Club| August 2023 - Present | 2 hrs. per wk.
Member | Billiards Club | | January 2024 - Present| 1 hrs. per wk.
Life Scout | Boy Scouts | June 2015 - March 2023 | 3 hrs. per wk.
Volunteer | St. Paul Catholic Church | June 2018 - August 2023 | 1 hrs. per wk.

WORK HISTORY

Worker | MSOE Kern Center | September 2025 – Present | 10 hrs. per wk.
Cook | Raised Grain Brewing Co. | June 2022 – September 2025 | 40 hrs. per wk.

INTERESTS

History | Weightlifting | Finance | Skiing | Reading | BJJ